

Ruhila Goswami

ruhila@ieee.org
RuhiRG

"Nothing exists for itself alone, but only in relation to other forms of life."
– Charles Darwin

Applicant Details

Current name	Ruhila Goswami
Previous name	Ruhila S.
Date of Birth	20.09.2001

Research Profile

Focus	Evolutionary divergence and dental variation in salmonids.
Interests	Population genetics; phenotypic-genotypic mapping; structural variation in genomes and selection; salmonid evolution; non-invasive methods.
Methods	Phylogenetic inference, genomic synteny, PCR-based sex determination, skeletal phenotyping, field sampling, and reproducible analysis in R and Python.

Education

2024–2026 **M.S. Biological Science**, *University of Iceland*, Reykjavík, *Iceland*
(Advisor: Prof. Arnar Pálsson)

Thesis: Evolutionary Divergence in Dental Variation and Sex Chromosome Architecture in Icelandic Brown Trout and Arctic Charr

2021–2024 **B.S. Biology (Major)**, *Indian Institute of Science Education and Research (IISER)*, Mohali, *India*
First Division with Honors

2020 **Intermediate (AISSCE)**, *Velammal Vidyalaya, Ayanabakkam*, Chennai, *India*
83.8% Central Board of Secondary Education (CBSE)

2017 **High School (AISSE)**, *Velammal Vidyalaya, Ayanabakkam*, Chennai, *India*
10.0 Cumulative Grade Point Average (CGPA) in Central Board of Secondary Education (CBSE)

Publications

Manuscripts

Under review rsx: a high-performance streaming toolkit for RAD-seq sex determination (with Rohit Goswami), *BMC Bioinformatics*

In preparation Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids

In preparation Concordant PCR and RAD-seq evidence localises the brown trout (*Salmo trutta*) sex locus to distal LG28 across Icelandic populations (with M. Lagunas and A. Pálsson)

Conference Proceedings

Rohit Goswami, Ruhila S , Amrita Goswami, Sonaly Goswami, and Debabrata Goswami. "Reproducible High Performance Computing Without Redundancy with Nix." In: *2022 Seventh International Conference on Parallel, Distributed and Grid Computing (PDGC)*. 2023.

Rohit Goswami and Ruhila S . "High Throughput Reproducible Literate Phylogenetic Analysis (accepted)." In: *2022 Seventh International Conference on Parallel, Distributed and Grid Computing (PDGC)*. 2023.

Preprints

Rohit Goswami, Ruhila S , Amrita Goswami, Sonaly Goswami, and Debabrata Goswami. *Unified Software Design Patterns for Simulated Annealing*. Feb. 6, 2023. arXiv: 2302.02811 [physics]. URL: http://arxiv.org/abs/2302.02811 (visited on 02/10/2023).

Science Communication

June 2026 **Hvernig þróast ósamhverfa í byggingu dýra? (How does asymmetry evolve in animal body structure?)**, *Vísindavefurinn (The Science Web), University of Iceland*, Arnar Pálsson, Ruhila Goswami
https://www.visindavefur.is/svar.php?id=88774

Presentations and Posters

Posters

April 2026 **Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids**, *Icelandic Ecological Society (Vistís)*, Ruhila Goswami

March 2026 **Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids**, *NOWPAS*, Ruhila Goswami

October 2025 **Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids**, *IceBio*, Ruhila Goswami

2025 **Diversity in craniofacial elements by populations and sexes in salmonids of Iceland**, *Icelandic Ecological Society (Vistís)*, Ruhila Goswami, Best Poster Award

November 2022 **Tracing Lineages of *Salmo salar* through Histone sequence data**, *BES Annual Meeting 2022*, Ruhila S. Goswami (née S.)

Oral Presentations

November 2022 **High Throughput Reproducible Literate Phylogenetic Analysis**, *IEEE PDGC 2022*, R. Goswami, Ruhila S. Goswami (née S.)

November 2022 **Reproducible High Performance Computing Without Redundancy with Nix**, *IEEE PDGC 2022*, R. Goswami, Ruhila S. Goswami (née S.), A. Goswami, S. Goswami and D. Goswami

November 2022 **Reproducible Literate Programming Workflows for Censored Data**, *IOP Machine Learning for Healthcare*, R. Goswami, R. S.

Honors and Awards

2025 **Best Poster Award**, *Icelandic Ecological Society (Vistís)*

Research Experience

Summer 2023 **Prof. Arnar Pálsson**, *University of Iceland*, Research Staff
Studied the synteny of histone genes in salmonids, focusing on gene arrangement and neighboring genomic regions.

Winter 2022 **Prof. Debabrata Goswami**, *IIT Kanpur*, Research Intern
Studied the basics of non-linear optics and applications of optical tweezers in cancer cell research and microscopy.

Winter 2022 **Dr. Nagma Parveen**, *IIT Kanpur*, Research Intern
Studied wet-lab methods for viral mechanisms under external stimuli and began work on a SARS-CoV-2 pseudovirus protocol using an HIV-1 lentiviral packaging system with a luciferase reporter.

Summer 2022 **Prof. Arnar Pálsson**, *University of Iceland*, Research Staff

Detailed analysis in a literate and reproducible manner for simulating a series of possible molecular evolutionary pathways for the *Salmonid* using phylogenetic trees. This involved the five steps on an HPC (High Performance Computer) with literate programming visualization in R:

- Data curation with NCBI databases
- Homology inference using similarity measures (BLAST)
- Multiple sequence alignment (MUSCLE5)
- Alignment trimming (Gblocks)
- Tree simulation with distance measures (BIONJ) and maximum likelihood approaches (RAxML-NG)

Project Report: Computational Primitives for Evolutionary Paths ($\approx$ 147 pages)

Summer 2021 **Dr. Lolitika Mandal**, *IISER Mohali*, Research Intern

Explored genetic tools for working with *Drosophila* from a wet-lab perspective of data collection and analysis.

Teaching Experience

Spring 2026 **LÍF401G Developmental Biology**, *University of Iceland*, Teaching Assistant

Course led by Prof. Adam Ray Smith. Assisted with embryo analysis for an 8 credit undergraduate developmental biology course.

Fall 2025 **LÍF541G Plant Physiology**, *University of Iceland*, Teaching Assistant

Course led by Prof. Victor Anthony Albert. Led student discussions in a 7.5 credit undergraduate course focused primarily on plant genetics.

Spring 2025 **LÍF403G Evolutionary Biology**, *University of Iceland*, Teaching Assistant

8 credit undergraduate course on evolutionary theory, natural selection, adaptation, and the history of life.

Research Skills

Genomics	Population genetics, genomic synteny, structural variation, PCR-based sex determination, non-invasive sampling	Phenotypes	Dental and craniofacial variation, bone extraction, otolith extraction, fish dissection
Evolutionary methods	Phylogenetic tree construction (distance, maximum likelihood, Bayesian), BLAST, MUSCLE5, Gblocks, RAxML-NG, BEAST2	Computing	R, Python, shell, Git, Snakemake, reproducible literate programming

Laboratory	DNA isolation and preservation, PCR, microscopy, *Drosophila* tissue dissection, cell culture
Field	Electrofishing, salmonid sampling and preservation

Service and Open Source

June-August 2024 **`pydSEAMSlib`: Enhanced Python bindings for d-SEAMS**, *Google Summer of Code*, Student Developer
Improved usability and accessibility of pySEAMS through documentation, installation refinements, and user-facing features.

June-August 2023 **`PYSEAMS`: Python bindings for d-SEAMS**, *Google Summer of Code*, Student Developer
Developed Python bindings for d-SEAMS, a molecular dynamics analysis engine for qualitative ice structure classification in confined and bulk systems. Refactored C++ code, removed Lua bindings, improved macOS support, and contributed to documentation, automated testing, and repository maintenance.

2022–present **IEEE P3173**, *IEEE Standards Committee*, Secretary
Supporting the drafting of the IEEE Standard for Endocrine Disrupting Chemical Hazard Labeling.

Affiliations

2024–present **Nordic Society Oikos**, Student Member

2024–present **Icelandic Ecological Society (Vistís)**, Student Member

2024–present **The Icelandic Biological Society (Líffræðigáttin)**, Student Member

2022–present **British Ecological Society**, Student Member

2022–present **Genetics Society, UK**, Undergraduate Member

2022–present **Genetics Society of America**, Undergraduate Member

Certifications

April 2024 **Biophotonics**, *IIT Kharagpur*, Distinction, 87%

April 2024 **Introduction to Professional Scientific communication**, *IIT Kanpur*, 79%

Sep 2022 **Applications of machine learning techniques in biology using WEKA**, *IIT Madras*, Distinction, 93%

Nov 2022 **Practical Python for beginners: a biochemist's guide**, *Biochemical Society, U.K.*